

The $\mathfrak{B}_3$ row

$\mathfrak{A}^{\dagger(1)} \times \bar{\mathfrak{B}}_3^{\dagger} \mathfrak{A}^{(1)}$ — checking the five terms, the divergences, and the p^+ integral

$$A \equiv (x-z)^2, \quad B \equiv (x-w)^2, \quad \beta \equiv \frac{k^+ B}{A}, \quad V^m \equiv k^+(x-w)^m - p^+(x-z)^m, \quad W \equiv k^+ B - p^+ A, \quad \mathcal{K} \equiv \frac{(x' - w') \cdot (y - z)}{(x' - w')^2 (y - z)^2}$$

Verdict. All five terms are right — every prefactor reproduces with ratio exactly 1, the argument map $\mathfrak{B}_3(k^+, z; p^+, -u) \rightarrow \mathfrak{B}_3(k^+, w; p^+, -z)$ is applied correctly in all five, and the colour structure is correct with f^{abd} *legitimately* an overall factor this time (§1–2). Four things need fixing (§3), one of which is substantive: the Θ 's on terms 4 and 5 are effectively swapped relative to the definition, and that decides whether a **double pole** cancels. On divergences: **no UV anywhere** in the transverse integrand; one unprotected infrared log, at $|z| \rightarrow \infty$ in term 3 (§4). The p^+ integral has **three** distinct singularities, and the important one is not a logarithm — it is a $\frac{1}{\Lambda}$ **power** divergence from the double pole at $p^+ = k^+$, plus an **on–contour pole** at $p^+ = \beta$ that needs a prescription (§5). Closed forms for terms I and III in §6.

1. The argument map and the prefactors

The vertex is defined as $\mathfrak{B}_{3ji}^{ba}(k^+, z; p^+, -u)$ and used as $\bar{\mathfrak{B}}_{3ji}^{\dagger ba}(k^+, w; p^+, -z)$, so $(x-z)_{\text{def}} \rightarrow (x-w)$ and $(x-u)_{\text{def}} \rightarrow (x-z)$. Every structure maps as you have it:

in the definition	in the row	your terms
$p^+(x-u)^2 - k^+(x-z)^2$	$p^+ A - k^+ B$	1, 2, 3 ✓
$k^+(x-z)^2 - p^+(x-u)^2$	$k^+ B - p^+ A = W$	4, 5 ✓
$(x-z) - (x-u)$	$z - w$	4, 5 ✓
$k^+(x-z) - p^+(x-u)$	V	5 ✓

With the bra $\frac{ig}{\sqrt{2\pi}\sqrt{k^+}}$, the ket's $\mathfrak{A}^{(1)}(p^+) \frac{ig}{\sqrt{2\pi}\sqrt{p^+}}$, and the conjugates of the three $\mathfrak{B}_3$ pieces:

term	quoted coefficient	computed / quoted
1, 2 (piece I)	$\frac{ig^4 f^{abd}}{8\pi^4} \frac{(p^+ + k^+)}{(k^+ - p^+)^2}$	1
3 (piece III)	$-\frac{ig^4 f^{abd}}{8\pi^4} \frac{1}{p^+ k^+}$	1
4 (piece II, $\frac{1}{(z-w)^2}$ half)	$\frac{ig^4 f^{abd}}{4\pi^4} \frac{1}{k^+(k^+ - p^+)}$	1
5 (piece II, $-\frac{p^+ k^+}{V^2}$ half)	$-\frac{ig^4 f^{abd}}{4\pi^4} \frac{p^+}{(k^+ - p^+)}$	1

All exact. Note the $\sqrt{p^+ k^+}$'s cancel correctly in each case, and in term 5 the $\frac{1}{k^+}$ of piece II combines with the $-p^+ k^+$ to leave $-\frac{p^+}{k^+ - p^+}$, which is what you have.

2. Colour — and this time f can come out

$\mathfrak{B}_3^{ba}$ carries $f^{abc} \rho^c(x)$; relabel $c \rightarrow d$. Barring rotates the source, and the $\mathfrak{A}^{(1)}$ half gives $[U^{bc}(z) - U^{bc}(y)] \rho^c(y)$, so

$$f^{abd} \left[U^{de}(x) U^{bc}(z) - U^{de}(x) U^{bc}(y) \right] \rho^e(x) \rho^c(y) \quad \checkmark$$

Here a is free and both b and d are contracted with Wilson lines, so f^{abd} is a genuine overall factor — unlike the $\mathfrak{A}^{(3)}$ row, where it was not. Your colour structure is correct as written.

3. Four fixes

1. **Term 3 has no measure.** Terms 1, 2, 4, 5 all carry $\int_{x,y,z,x'}$; term 3 does not, although x, y, z, x' all appear in it.
2. **Bra transpose.** The row gives $[U^{ab'}(x') - U^{b'a}(w')]$; you wrote $[U^{ab'}(x') - U^{ab'}(w')]$. Same issue as in the previous two rows — pick one ordering.
3. **The $\mathfrak{B}_3$ definition has an unbalanced bracket.** The $\backslashleft[$ opening $[\frac{1}{((x-z)-(x-u))^2}\Theta(\dots) - \frac{p^+k^+}{(\dots)^2}$ never closes, so piece III is typeset *inside* piece II's bracket. The five terms you extracted show you meant them separate, but the source needs fixing.
4. **The Θ 's on terms 4 and 5 are swapped** relative to the definition. The definition puts $\Theta(p^+ - k^+ - \Lambda)$ on the $\frac{1}{(z-w)^2}$ half (so $p^+ > k^+ + \Lambda$) and *none* on the $-\frac{p^+k^+}{V^2}$ half (full range). You have term 4 on $[\Lambda, k^+ - \Lambda]$ and term 5 on $[k^+ + \Lambda, \vee]$. This is not cosmetic — §5 shows it decides whether a double pole cancels.

4. Transverse divergences

4.1 No ultraviolet divergence

Every subset of $\{x, y, z, x', w, w'\}$ of size ≥ 2 was collapsed at scale ρ against the measure $\rho^{2(n-1)}\frac{d\rho}{\rho}$, for all four structures. **Not one region diverges.** The two that look dangerous both turn out safe:

- $x, z, w \rightarrow$ a point, where $p^+A - k^+B \rightarrow 0$ like ρ^2 : integrand ρ^{-2} against a three-point measure $\rho^3 d\rho$ — convergent.
- $V = 0$, i.e. $k^+(x - w) = p^+(x - z)$ as *vectors*, at $x_* = \frac{k^+w - p^+z}{k^+ - p^+}$. Term 5 has $\frac{1}{V^2}$ there, but its numerator V^m vanishes linearly, so the integrand goes as $\frac{1}{\rho}$ against $\rho d\rho$ — integrable.

4.2 The large-distance end

Angular average $\times R^2$ for the bare integrand; a flat column is a logarithm:

variable	I	III	IIa	IIb
$ x \rightarrow \infty$	8.0488 flat		-4.3170 flat	conv
$ y \rightarrow \infty$		conv	conv	conv
$ z \rightarrow \infty$		conv	-2.4710 flat	conv
$ x' \rightarrow \infty$		conv	conv	conv
$ w \rightarrow \infty$		5.0708 flat	conv	5.6865 flat
$ w' \rightarrow \infty$		conv	conv	-2.1040 flat

Flat to five digits over four decades. Now sort them:

- $|w|$ carries the phase $e^{+ik \cdot w}$, which gives the usual Bessel $R^{-1/2}$ damping — convergent once the phase is restored. (Same mechanism as the two- $\mathfrak{C}$ row.)
- $|x|$ is a source position: the integral is cut off by the support of $\rho(x)$.
- $|z|$ **in term 3 is the one that is not protected.** z is a genuine integration variable, it carries no phase, and the term survives. Here $\frac{(x-z)(y-z)}{(x-z)^2(y-z)^2} \sim |z|^{-2}$ while $\frac{k^+B + p^+A}{p^+A - k^+B} \rightarrow 1$, so the integrand is $|z|^{-2}$ against $|z|^2 \frac{d|z|}{|z|}$.

So: one genuine infrared log, at $|z| \rightarrow \infty$ in term 3, of exactly the character found in the previous two rows. Regulate it the same way — $\int d^{2-2\epsilon}z$ with $\epsilon > 0$ — giving a $\frac{1}{\epsilon_{\text{IR}}}$ pole. There is no UV pole to pair it with in this row.

5. The p^+ integral has three singularities

5.1 A double pole at $p^+ = k^+$ — and it does not fully cancel

Residues $\lim_{p^+ \rightarrow k^+} (k^+ - p^+)^2 \times (\text{term})$, computed with the quoted prefactors, converge cleanly (8 digits, from both sides):

term	I	III	IIa	IIb
residue	+0.00218023	0	+0.00218023	−0.00218023

$$\text{residue(I)} = \frac{1}{4\pi^4} \frac{\mathcal{K}}{A - B} \quad (\text{predicted, matches to 8 digits}).$$

IIa and IIb cancel each other exactly. That is clearly why piece II is written as a *difference*: at $p^+ = k^+$ one has $V \rightarrow k^+(z - w)$, so $\frac{p^+k^+}{V^2} \rightarrow \frac{1}{(z-w)^2}$ and the two halves coincide. **But term I's double pole has nothing to cancel against.**

Consequently the Θ strip $|p^+ - k^+| > \Lambda$ leaves a **power** divergence:

$$\int_{\Lambda}^{k^+-\Lambda} + \int_{k^++\Lambda}^{\vee} \frac{dp^+}{(k^+ - p^+)^2} = \frac{2}{\Lambda} - \frac{1}{k^+} - \frac{1}{\vee - k^+} \Rightarrow \boxed{\frac{4}{\Lambda(A - B)}}$$

This is $\frac{1}{\Lambda}$, **not** $\log \frac{1}{\Lambda}$. It cannot be absorbed by rapidity evolution and it will not cancel inside this row. Two things follow. First, with *your* ranges the IIa/IIb cancellation happens above k^+ (terms 1 and 5) but below k^+ terms 2 and 4 **add**, doubling the residue — so the range assignment is not self-consistent. With the *definition's* Θ 's it cancels below and survives above. Neither choice makes it cancel on both sides. Second, either way term I's own double pole survives. Resolve fix 4 of §3 first; if the $\frac{1}{\Lambda}$ still stands, it has to cancel against a companion row.

5.2 An on-contour pole at $p^+ = \beta$

Every term contains $p^+A - k^+B$, which vanishes at

$$p_*^+ = \beta = k^+ \frac{(x - w)^2}{(x - z)^2},$$

a **transverse-dependent** point that lies inside $[\Lambda, \vee]$ for generic configurations. This is a simple pole *on* the integration contour, so the p^+ integral is not defined until you supply the prescription. It is an energy denominator, so the $+i\epsilon$ of the propagator gives

$$\frac{1}{p^+A - k^+B \pm i\epsilon} = \text{PV} \frac{1}{p^+A - k^+B} \mp i\pi \delta(p^+A - k^+B).$$

The δ piece is **not** optional: every prefactor here carries an explicit i , so $i \times (\mp i\pi)$ is **real** and contributes to the cross section. Physically it is the on-shell intermediate state — a genuine real-emission contribution. Dropping it, or taking a bare principal value, loses a real part of the answer.

5.3 Soft poles at $p^+ \rightarrow 0$

Term 3 carries $\frac{1}{p^+k^+}$ and term 4's bracket carries $-\frac{k^+}{p^+}$. These are the usual rapidity poles and are handled by the $[\frac{1}{p^+}]_+$ prescription as before.

6. The p^+ integrals in closed form

Both primitives verified symbolically ($\frac{dF}{dp^+} - \text{integrand} = 0$ exactly) and against principal-value quadrature to six digits.

6.1 Term I

$$F_I(p^+) = -\frac{2}{(A - B)(p^+ - k^+)} + \frac{A + B}{k^+(A - B)^2} \log \left| \frac{p^+ - \beta}{p^+ - k^+} \right|$$

$$\mathcal{T}_I = \frac{4}{\Lambda(A - B)} - \frac{2}{A - B} \left[\frac{1}{k^+} + \frac{1}{\vee - k^+} \right] + \frac{A + B}{k^+(A - B)^2} \left[\log \frac{A}{B} + \log \frac{\vee - \beta}{\vee - k^+} \right] \mp i\pi \frac{A + B}{k^+(A - B)^2}$$

There is no $\log \Lambda$ at all — the logarithms from the two strip edges cancel against each other, leaving only the $\frac{1}{\Lambda}$ power term. As $\vee \rightarrow \infty$ the second logarithm vanishes and

$$\mathcal{T}_I \longrightarrow \frac{4}{\Lambda(A-B)} - \frac{2}{k^+(A-B)} + \frac{A+B}{k^+(A-B)^2} \log \frac{A}{B} \mp i\pi \frac{A+B}{k^+(A-B)^2}.$$

6.2 Term III

Partial fractions give $\frac{k^+B+p^+A}{p^+(p^+-\beta)} = -\frac{A}{p^+} + \frac{2A}{p^+-\beta}$, so

$$\mathcal{T}_{III} = \frac{1}{k^+} \left[-\log \frac{\vee}{\Lambda} + 2 \log \left| \frac{\vee - \beta}{\Lambda - \beta} \right| \right] \xrightarrow{\Lambda \rightarrow 0, \vee \rightarrow \infty} \frac{1}{k^+} \log \frac{\vee \Lambda A^2}{k^{+2} B^2} \mp \frac{2i\pi}{k^+}$$

Note the unusual shape: the rapidity logarithm and the $\vee$ logarithm appear as a **product** $\log(\vee \Lambda)$, not a ratio. The residue 2 at $p^+ = \beta$ (against -1 at $p^+ = 0$) is what produces it.

6.3 Terms IIa and IIb

Here p^+ sits *inside* the tensor contraction — both V^m and W depend on it — so there is no single scalar master. What is fixed is the pole content: the $\frac{1}{(k^+-p^+)^2}$ pieces cancel between the two (§5.1), the remaining poles are the $p^+ = \beta$ pole of §5.2 (from W), the $p^+ \rightarrow 0$ pole of §5.3 (from the $-\frac{k^+}{p^+}$ in the bracket), and the V^2 zero in IIb, which §4.1 shows is integrable. Expand the $\delta_{ik}\delta_{jm}$, $\delta_{jk}\delta_{im}$, $\delta_{ij}\delta_{km}$ structures separately and each piece reduces to the masters above with A, B replaced by the appropriate contractions.

7. What to do next

1. Fix the four items in §3 — especially the Θ assignment, since §5.1 shows it changes whether a double pole cancels.
2. Decide the $i\epsilon$ at $p^+ = \beta$ from the light-cone energy denominator and keep the δ -function piece; it is real here.
3. Regulate the $|z| \rightarrow \infty \log$ in term 3 in $d = 2 - 2\epsilon$.
4. Then track the $\frac{4}{\Lambda(A-B)}$: it is a power divergence, so it must cancel against another row, not be absorbed.

zint/src/row11_pref.py (§1–2), row11_sing.py, row11_res.py (§4.1, §5.1), row11_far.py (§4.2), row11_pint.py (§6) — every number above.